

Program : Diploma in Information Technology	
Course Code : 6268	Course Title: Smart Systems Lab
Semester : 6	Credits: 2.5
Course Category: Program Core	
Periods per week: 4 (L:1 T:0 P:3)	Periods per semester: 60

Course Objectives:

- To familiarize students with Arduino as IDE, programming language & platform.
- To develop skills to design and implement various smart system applications.

Course Prerequisites:

Topic /Description	Course Code	Course Title	Semester
Knowledge of basic Electronics		Fundamentals of Electrical & Electronics Engineering Lab	2
Basic knowledge of problem solving through programming		Problem solving and Programming	2

Course Outcomes

On completion of the course, students will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO 1	Illustrate Arduino board and its operations	13	Understanding
CO 2	Make use of Arduino IDE to develop programs for lighting LED	15	Applying
CO 3	Make use of Arduino IDE and Arduino Board to develop programs for LCD displays and seven segment display	15	Applying
CO 4	Make use of Arduino IDE and Arduino Board to develop programs for DC motor and servo motor	14	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2			2			
CO2	3			3			
CO3	3	3	3	3			3
CO4	3						3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate Arduino board and its operations		
M1.01	Introduction to Arduino IDE and different types of Arduino Board	4	Understanding
M1.02	Syntax for writing Arduino IDE	4	Applying
M1.03	Blink LED in Arduino board	5	Applying
CO2	Make use of Arduino IDE to develop programs for lighting LED		
M2.01	Blinking of LED, Fading of LED, Circling of LED (for loop), traffic light system	9	Applying
M2.02	Changing the brightness of LED using potentiometer	6	Applying
	Lab Exam – I	1.5	
CO3	Make use of Arduino IDE and Arduino Board to develop programs for LCD displays and seven segment display.		
M3.01	Display a message in LCD screen, scrolling of text in LCD screen	5	Applying
M3.02	Simple automatic count down and count up (for loop) for seven segment display	5	Applying
M3.03	Increment and decrement a number by using push button for seven segment display	5	Applying

CO4	Make use of Arduino IDE and Arduino Board to develop programs for DC motor and servo motor		
M4.01	Controlling a DC motor	7	Applying
M4.02	Direction control of servo motor	7	Applying
	Lab Exam – II	1.5	

*****Suggested Open-ended Experiments:**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open-ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

1. Create a Bluetooth controlled toy car using Arduino.

Text / Reference:

T/R	Book Title / Author
T1	21 IOT experiments by Yashavant Kanetkar, ShrirangKorde

Online Resources:

Sl.No	Website Link
1	https://www.arduino.cc/en/Main/Software